

Reflection Template

Use Case	UC-10
Use Case Name	Assign Task to User
Requirement IDs	FR11, FR20, FR22, FR23
Matching Feature Comparison Sets	CS-21, CS-22
Implementation Order	Kotlin first -> Java second

Feature Relevance

Feature/s:

Null safety, extension functions

Did the feature/s occur naturally in this/these context/s?

Yes, assignedUser is nullable in Kotlin (User?), setting it to a value that isnt null after the assign call felt natural. The extension function was reused again for task and user lookup. In Java for same logic setters and explicit constructor extension was used.

Qualitative Ratings (1-5)

- 1 = very poor/very difficult
- 2 = rather poor/difficult
- 3 = neutral/moderate
- 4 = good/easy
- 5 = very good/very easy

Criteria	Java	Kotlin
Readability	4	5
Compactness of logic	3	4
Perceived implementation effort	4	5

Justifications

Readability:

Both implementations are equally readable. The extension function reads naturally in both, findByIdOrThro clearly communicates intent in Kotlin, RepositoryUtils.findByIdOrThrow does the same in Java.

Compactness of logic:

Both are compact. No meaningful difference for this UC.

Implementation effort:

Straightforward in both languages. No new patterns were introduced.

Observed structural differences

Extension function vs RepositoryUtils, same consistent pattern as previous use cases. Kotlin calls read more naturally on the repository object.

Both request DTOs are single-line, no meaningful difference.

Final comparative summary

Here confirming existing patterns without introducing new ones. No feature-relevant differences emerged in this use case.